EXERCISE 11 – IMPLEMENTATION OF A FUZZY INFERENCE SYSTEM

AIM

To implement a Fuzzy Inference System for temperature-based fan-speed control.

PROCEDURE

1. Import NumPy, scikit-fuzzy, and Matplotlib libraries.

2. Define the input temperature range and output fan-speed range.

3. Create fuzzy membership functions for temperature and fan speed.

4. Plot the membership functions and choose the input temperature.

5. Fuzzify the input, apply the fuzzy rules, and calculate the crisp fan-speed output using defuzzification.

6. Run the program and observe the displayed output.

PROGRAM

```
import numpy as np
import skfuzzy as fuzz
import matplotlib.pyplot as plt

# Step 1: Define fuzzy variables
temperature = np.arange(0, 41, 1)  # Temperature: 0 to 40 degrees Celsius
fan_speed = np.arange(0, 11, 1)    # Fan speed: 0 to 10

# Step 2: Define fuzzy membership functions
# Temperature membership functions: Low, Medium, High
temp_low = fuzz.trimf(temperature, [0, 0, 20])
temp_medium = fuzz.trimf(temperature, [10, 20, 30])
# Note: the source file does not provide a definition for temp_high.

# Fan speed membership functions: Low, Medium, High
fan_low = fuzz.trimf(fan_speed, [0, 0, 5])
fan_medium = fuzz.trimf(fan_speed, [2, 5, 8])
fan_high = fuzz.trimf(fan_speed, [5, 10, 10])

# Step 3: Visualize the membership functions
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.plot(temperature, temp_low, label='Low')
plt.plot(temperature, temp_medium, label='Medium')
plt.plot(temperature, temp_high, label='High')
plt.title("Temperature Membership Functions")
plt.xlabel("Temperature (°C)")
plt.ylabel("Membership Degree")
plt.legend()

plt.subplot(2, 1, 2)
plt.plot(fan_speed, fan_low, label='Low')
plt.plot(fan_speed, fan_medium, label='Medium')
plt.plot(fan_speed, fan_high, label='High')
plt.title("Fan Speed Membership Functions")
```

```
plt.xlabel("Fan Speed")
plt.ylabel("Membership Degree")
plt.legend()
plt.tight_layout()
plt.show()

# Step 4: Fuzzification
temperature_input = 28
temp_low_level = fuzz.interp_membership(temperature, temp_low, temperature_
temp_medium_level = fuzz.interp_membership(
    temperature, temp_medium, temperature_input)
temp_high_level = fuzz.interp_membership(temperature, temp_high, temperatu

# Step 5: Apply fuzzy rules
# Rule 1: If temperature is Low, fan speed is Low
# Rule 2: If temperature is Medium, fan speed is Medium
# Rule 3: If temperature is High, fan speed is High
fan_low_level = temp_low_level
fan_medium_level = temp_medium_level
fan_high_level = temp_high_level

# Step 6: Defuzzification
fan_output = fuzz.defuzz(
    fan_speed,
    fan_low * fan_low_level + fan_medium * fan_medium_level
    + fan_high * fan_high_level,
    'centroid'
)

# Step 7: Print the output
print(f"Temperature: {temperature_input}°C")
print(f"Fuzzified fan speed: {fan_output:.2f}")
```

OUTPUT

```
The source file does not specify a fixed output value.
Note: temp_high is used in the source code but its membership function is n
The program needs that definition before it can run successfully.
```